FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2022

PART TEST – III

PAPER –1

TEST DATE: 19-12-2021

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

Section – A *(01 – 04, 20 – 23, 39 – 42)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(05 –10, 24 – 29, 43 – 48)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B *(11 – 13, 30 – 32, 49 – 51)*: This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

Section – C *(14 – 19, 33 – 38, 52 – 57)*: This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

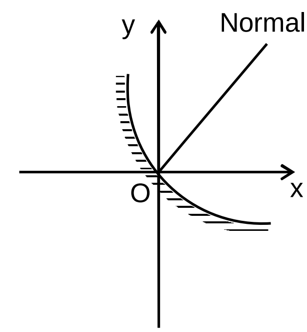
Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. A ray is incident on a curved spherical mirror of radius of curvatue 10 cm along a vector $\left(-\hat{i}-2\hat{j}+2\hat{k}\right)$. The direction of normal at the point of incidence is along the vector $\left(3\hat{i}+4\hat{j}\right)$ as shown in the figure. The unit vector along the reflected ray is



(A) $\frac{41}{75}\hat{i}+\frac{38}{75}\hat{j}-\frac{2}{3}\hat{k}$

(B) $\frac{41}{75}\hat{i}-\frac{38}{75}\hat{j}+\frac{2}{3}\hat{k}$

(C) $\frac{41}{75}\hat{i}+\frac{38}{75}\hat{j}+\frac{2}{3}\hat{k}$

(D) $-\frac{41}{75}\hat{i}-\frac{38}{75}\hat{j}-\frac{2}{3}\hat{k}$

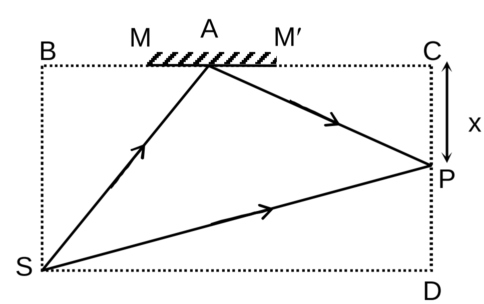
Ans. C

Sol. $\hat{e}_r=\hat{e}_i-2(\hat{e}_i\cdot\hat{n})\hat{n}$

$=\left(\frac{-\hat{i}-2\hat{j}+2\hat{k}}{3}\right)+2\left(\frac{11}{15}\right)\left(\frac{3\hat{i}+4\hat{j}}{5}\right)$

$=\frac{41}{75}\hat{i}+\frac{38}{75}\hat{j}+\frac{2}{3}\hat{k}$

2. A monochromatic point source 'S' of wavelength 'λ' is kept in front of a plane mirror MM′. Interference is observed at point 'P' of two waves SAP and SP. It is given that AB = $\sqrt{14}x$, $AC=\sqrt{8}x$, $BS=\sqrt{2}x$, $SD=\sqrt{2(11+\sqrt{2})}x$. The minimum value of 'x' for which the maximum intesnity is obtained at point P after interference is



(A) λ

(B) $\frac{3}{4}\lambda$

(C) $\frac{\lambda}{2}$

(D) $\frac{\lambda}{4}$

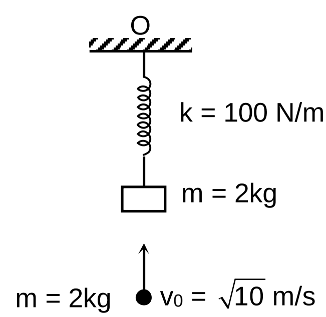
Ans. D

Sol. Path difference, $\Delta r = \left(SA + AP + \frac{\lambda}{2}\right) - SP = 2x + \frac{\lambda}{2}$

For maxima, $2x + \frac{\lambda}{2} = n\lambda$

$x = \lambda/4$

3. A massless spring of spring constant k = 100 N/m hangs in a vertical plane from point 'O', its other end is attached with a block of mass m = 2kg and system is in equilibrium. Now another small spherical ball of same mass moving with a velocity $v_0 = \sqrt{10}$ m/s collides with the block and gets stuck to it as shown in the figure. The new amplitude of oscillation will be



(A) $\frac{\sqrt{34}}{5}$ m

(B) $\frac{\sqrt{14}}{10}$ m

(C) $\frac{\sqrt{17}}{2}$ m

(D) $\frac{\sqrt{13}}{4}$ m

Ans. B

Sol. $mv_0 = (m + m)v$

$v = \frac{v_0}{2}$ …(i)

$v = \omega\sqrt{A^2 - x^2}$

$\frac{v_0}{2} = \sqrt{\frac{k}{2m}}\sqrt{A^2 - \left(\frac{mg}{k}\right)^2}$

$A = \sqrt{\frac{mv_0^2}{2k} + \frac{m^2g^2}{k^2}} = \frac{\sqrt{14}}{10}$ m

4. In a setup of photoelectric experiment, a light of intensity 384 W/m^2 is used which is equally distributed among three wavelengths 620 nm, 496 nm and 310 nm. The light is incident perpendiculary on a surface of area 0.01 cm^2, having work function of 2.25 eV. Assuming there is no loss of light and efficiency of emission of electrons from the surface is 50%. The photocurrent in the circuit is

(A) 83.2×10^{-6} amp

(B) 166.4×10^{-6} amp

(C) 20.8×10^{-6} amp

(D) 41.6×10^{-6} amp

Ans. D

Sol. Energy of light

$$E_1 = \frac{1240}{620} = 2 \text{ eV}$$

$$E_2 = \frac{1240}{496} = 2.5 \text{ eV}$$

$$E_3 = \frac{1240}{310} = 4 \text{ eV}$$

No. of photons capable to emit electorns per second

$$= \frac{\left(\frac{384}{3}\right)\times 10^{-6}}{2.5\text{ e}} + \frac{\left(\frac{384}{3}\right)\times 10^{-6}}{4\text{e}} = \frac{83.2\times 10^{-6}}{\text{e}}$$

No. of electrons emitted per second = $\frac{83.2\times 10^{-6}}{2\times \text{e}}$

Photo current = ne = 41.6×10^{-6} ampere

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

5. h, G, S, η, μ_0 and ε_0 are the plank constant. Gravitational constant, surface tension, coefficient of viscocity, permeability and permittivity of free space respectively. [M], [L], [T] and [A] denote the fundamental dimension of mass, length, time and current respectively. Regarding the dimension of some physical quantity choose the correct option(s).

(A) Dimension of $\frac{hG}{\eta^2}$ is $\left[M^{-1}L^{7}T^{-1}\right]$

(B) Dimension of $\frac{\mu_0}{Sh}$ is $\left[M^{-1}L^{-1}T^{-1}A^{-2}\right]$

(C) Dimension of $\frac{hG}{\varepsilon_0}$ is $\left[ML^{8}T^{-7}A^{-2}\right]$

(D) Dimension of $\frac{S\varepsilon_0}{\eta}$ is $\left[M^{-1}L^{-2}T^{3}A^{2}\right]$

Ans. C, D

Sol. Dimension of h is $\left[ML^2T^{-1}\right]$

Dimension of G is $\left[M^{-1}L^3T^{-2}\right]$

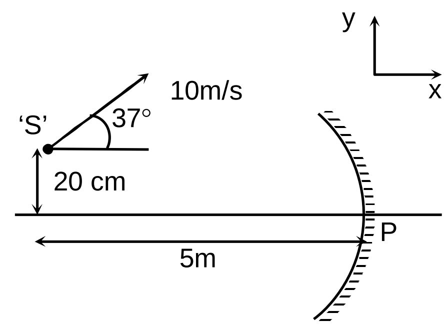
Dimension of S is $\left[MT^{-2}\right]$

Dimension of η is $\left[ML^{-1}T^{-1}\right]$

Dimension of μ_0 is $\left[MLT^{-2}A^{-2}\right]$

Dimension of ε_0 is $\left[M^{-1}L^{-3}T^4A^2\right]$

6. A concave mirror of focal length 3 meter is moving in the x-y plane with a velocity $(4\hat{i}+4\hat{j})$ m/s. A point source 'S' is also moving in the x-y plane with a velocity 10 m/s as shown in the figure. At a given instant, the object is at 5 m along the negative x-axis from pole and 20 cm along positive y-axis form the pole. The pole is assumed to be origin at this instant. Assuming paraxial approximation, choose the correct option(s).



(A) Velocity of image with respect to mirror at the same instant is $\left(-9\hat{i}-\frac{18}{5}\hat{j}\right)$ m/s

(B) Velocity of image with respect to mirror at the same instant is $\left(-18\hat{i}-\frac{51}{5}\hat{j}\right)$ m/s

(C) Velocity of image with respect to ground at the same instant is $\left(-5\hat{i}+\frac{2}{5}\hat{j}\right)$ m/s

(D) Velocity of image with respect to source at the same instant is $\left(-13\hat{i}-\frac{28}{5}\hat{j}\right)$ m/s

Ans. A, C, D

Sol. $m=\frac{f}{f-u}$ …(i)

$\frac{h_{\hat{i}}}{h_0}=\frac{f}{f-u}$

$h_{\hat{i}}=h_0\left(\frac{f}{f-u}\right)$ …(ii)

$v=u\frac{f}{u-f}$ …(iii)

Differentiate equation (ii) and (iii) with respect time then we will get velocity of image.

7. Choose the correct statement(s).

(A) A particle of mass m at rest disintegrates into two particles of masses m_1 and m_2. Let λ_1 and λ_2 are their de-Broglie wavelengths respectively. It is given that $m_1 > m_2$ then $\lambda_1 > \lambda_2$

(B) The process of photoelectric emission is different to that of thermionic emission.

(C) Two radioactive samples A and B initially contain same number of nuclei. Sample A has half life of 1 year and B has half life of 2 years. At the end of 2 years, both samples have the same activity.

(D) When ultraviolet light is incident on a photocell, stopping potential is V_0 and the maximum kinetic energy of the photoelectrons is K_{max}. When the ultraviolet light is replaced by x-ray. K_{max} increases but V_0 decreases.

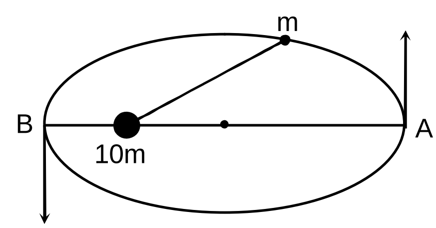
Ans. B, C

Sol. Use conservation of momentum

$$\lambda = \frac{h}{p}$$

Activity = λN

8. A satellite of mass 'm' is moving around a stationary planet of mass 10 m in an elliptical orbit of eccentricity e = 1/3 and semi-major axis 'a' as shown in the figure. Now choose the correct option(s).



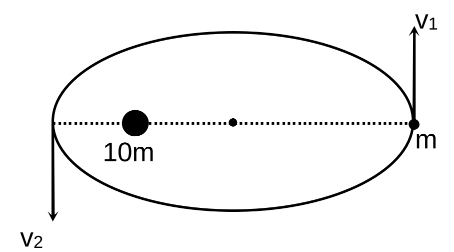
(A) The velocities of satellite at perihelion and aphelion are $2\sqrt{\frac{5Gm}{a}}$ and $\sqrt{\frac{5Gm}{a}}$ respectively.

(B) The radius of curvature of the path of the satellite at perihelion is $\left(\frac{8}{9}a\right)$

(C) The radius of curvature of the path of the satellite at perihelion is $\left(\frac{2}{3}a\right)$

(D) Total time taken by the satellite in one complete revolution is $\frac{2\pi a^{3/2}}{\sqrt{Gm}}$

Ans. A, B



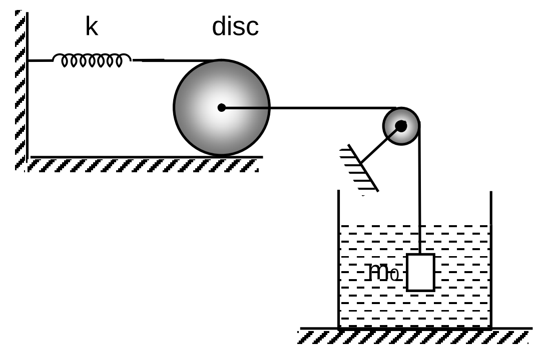
Sol. $mv_2\left(\frac{2a}{3}\right) = mv_1\left(\frac{4a}{3}\right)$ …(i)

$\frac{1}{2}mv_2^2 - \frac{Gm(10m)}{(2a/3)} = \frac{1}{2}mv_1^2 - \frac{Gm(10m)}{4a/3}$ …(ii)

$v_1 = \sqrt{\frac{5Gm}{a}}, v_2 = 2\sqrt{\frac{5Gm}{a}}$

$\frac{Gm(10m)}{\left(\frac{2a}{3}\right)^2} = \frac{mv_2^2}{r_2}, \ r_2 = \frac{8a}{9}$

9. A disc of mass 'm' and radius 'R' is kept on a rough horizontal surface in a vertical plane. A massless spring of spring constant 'k' is attached at the top of disc. The centre of disc is connected with a block of mass 'm_0' which is submerged in water as shown in the figure. The relative density of block is 3. There is sufficient friction between the disc and the surface to prevent slipping. Take m_0 = 2m and g′ is acceleration due to gravity.



(A) The elongation in the spring in equilibrium is $\frac{2mg}{3k}$

(B) The elongation in the spring in equilibrium is $\frac{mg}{3k}$

(C) If the disc is slightly displaced from its equilibrium position and released then its angular frequency of oscillations will be $\sqrt{\frac{8k}{3m}}$

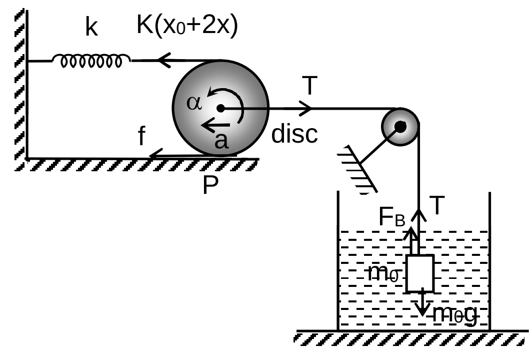
(D) If the disc is slightly displaced from its equilibrium position and released then its angular frequency of oscillations will be $\sqrt{\frac{8k}{7m}}$

Ans. A, D

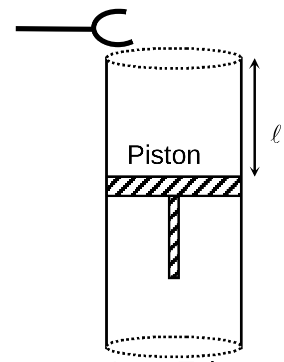
Sol. At equilibrium
$m_0g = T_0 + F_B$ …(i)
$T_0 = kx_0 + f_0$ …(ii)
$kx_0.R = f_0R$ …(iii)
$2kx_0 = T_0$ …(iv)
$\Rightarrow x_0 = \frac{2mg}{3k}$
Now if disc is slightly displaced

$T - m_0 g + F_B = m_0 a$ …(v)
$k(x_0 + 2x) + f - T = ma$ …(vi)
$k(x_0 + 2x).2R - TR = I_P\alpha$ …(vii)
On solving

$$a = \frac{4k}{\left(\frac{I_P}{R^2} + m_0\right)} x = \frac{8k}{7m}.x$$



10. A standing wave is formed in a long tube by a tuning fork of frequency 500 Hz kept at its open end and other end is closed by a movable piston as shown in the figure. A resonance is produced when ℓ = 18.0 cm, 55.5 cm and 93.0 cm respectivley. The temperature of air is maintained at 77°C and molecular weight of air is 28.8×10^{-3} kg/mole. Choose the correct option(s).



(A) The minimum distance between node and antinode is 18.75 cm

(B) The speed of wave in the tube is approximately 375 m/s.

(C) The adiabatic coefficient of air inside the tube is approximately 1.39

(D) There is an end correction of 0.75 cm

Ans. A, B, C, D

Sol. $\frac{\lambda}{2} = 37.5$ cm , λ = 75.0 cm

So, $\frac{\lambda}{4} = 18.75$ cm

$\Rightarrow v = f\lambda = 375$ m/s

$\Rightarrow v = \sqrt{\frac{\gamma RT}{M}}$

$\gamma = 1.39$

$\Rightarrow$ end correction = 18.75 – 18.00 = 0.75

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

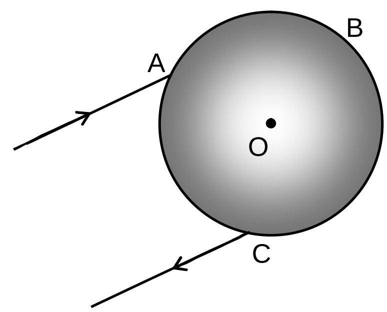
11. Consider a planet whose density is half of the earth and its radius is double of the earth. What will be the time period (in sec) of oscillation of the pendulum on this planet if it is a second pendulum of the earth.

Ans. 2

Sol. $T = 2\pi\sqrt{\frac{L}{g}}$

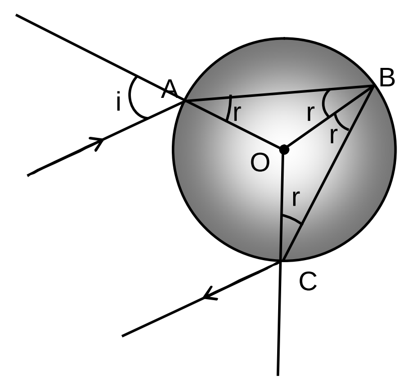
$T = k\sqrt{\frac{1}{\rho R_e}}$

12. A sphere of refractive index $\sqrt{2}$ is kept in a space whose centre is O. A ray is incident at point 'A' of the sphere and after refraction at A it is internally reflected at point 'B' then refracted at 'C' and finally comes out from the sphere. It is given that the total deviation produced in the incident ray is minimum, if the angle of incidence is given by $i = \sin^{-1}\left(\sqrt{\frac{2}{\lambda}}\right)$, then find the value of λ.



Ans. 3

Sol. Net deviation



$\delta = (i - r) + (\pi - 2r) + (i - r)$

$= 2(i - r) + (\pi - 2r)$

For minimum deviation $\frac{d\delta}{di} = 0$

$\frac{dr}{di} = \frac{1}{2}$ …(i)

Snell's law

$\sin i = \mu \sin r$

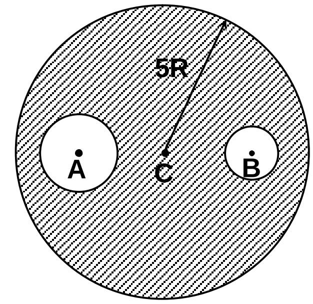
$\frac{dr}{di} = \frac{\cos i}{\mu \cos r}$ …(ii)

On solving

$\sin^2 i = \frac{4}{3}\left[1 - \frac{\mu^2}{4}\right]$

$\sin i = \frac{2}{3}$

13. A liquid drop of radius 5R is kept in the open atmosphere of pressure P_0. C is the centre of the liquid drop. Two air bubbles are formed inside the liquid drop centred at A and B of radii 2R and R respectively. Let the gauge pressure at A, B and C are P_A, P_B and P_C respectively. The ratio of $\frac{P_A P_B}{P_C^2}$ is equal to 3λ. Find the value of λ.



Ans. 7

Sol. $P'_C = P_0 + \frac{2S}{5R}$

$P_C = P'_C - P_0 = \frac{2S}{5R}$

$P'_A = P'_C + \frac{2S}{2R} = P_0 + \frac{7S}{5R}$

$P_A = P'_A - P_0 = \frac{7S}{5R}$

$P'_B = P'_C + \frac{2S}{R} = P_0 + \frac{12S}{5R}$

$P_B = P'_B - P_0 = \frac{12S}{5R}$

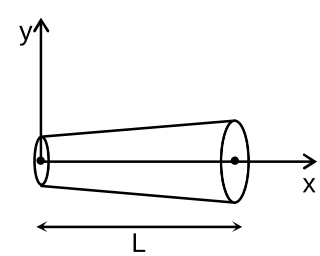
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

A rope of length 'L' is kept along x-axis such that its one end is at the origion as shown in the figure. An external force is applied on rope such that restoring force developed in the rope is given by $F_0\left(1+\frac{x}{L}\right)$, where F_0 is constant. The young's modulus of the rope is given by $Y_0\left(1+\frac{x}{L}\right)^{-1}$ and cross-sectional area is $A_0\left(1+\frac{x}{L}\right)$, where Y_0 and A_0 are constant. The elongation in the rope is $\Delta\ell$ mm and elastic potential energy stored in the rope is λ joule. It is given that F_0 = 1kN, $Y_0 = 2 \times 10^{11}$ N/m^2, A_0 = 5 mm^2 and L = 1m.



14. The value of $\Delta\ell$ is …………….

Ans. 1.50

15. The value of λ is ……………

Ans. 1.17

Sol. (Q.14-15)

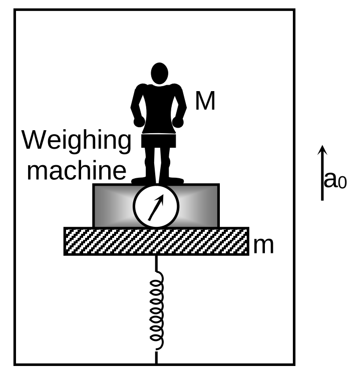
Net elongation = $\int_0^L \frac{F_x}{A_x Y_x} dx = \frac{F_0}{Y_0 A_0}\int_0^L \left(1+\frac{x}{L}\right) dx = \frac{3F_0 L}{2Y_0 A_0}$

Total potential energy = $\frac{1}{2}\int_0^L \frac{F_x^2}{A_x Y_x} dx = \frac{F_0^2}{2Y_0 A_0}\int_0^L \left(1+\frac{x}{L}\right)^2 dx = \frac{7}{6}\frac{F_0^2 L}{Y_0 A_0}$

Question Stem for Question Nos. 16 and 17

Question Stem

An elevator is moving vertically upward with an acceleration a_0 = 4 m/s^2. A man of mass M = 60 kg is standing on a massless weighing machine which is kept on a platform of mass m = 10 kg. Platform is connected with a spring as shown in the figure. Platform oscillates with a frequency of 0.5 sec^{-1} and amplitude 0.2 m. N_0 is the maximum reading of the weighing machine in kilogram and x_0 is the additional compression (in meter) in the spring from equilibrium position when the reading was 90 kg. (Take π^2 = 10, g = 10 m/s^2)



16. The value of N_0 is ……………..

Ans. 96.00

17. The value of x_0 is …………

Ans. 0.10

Sol. (Q.16-17)

$N_{max} = Mg + Ma_0 + M\omega^2 A$

= 960 Newton

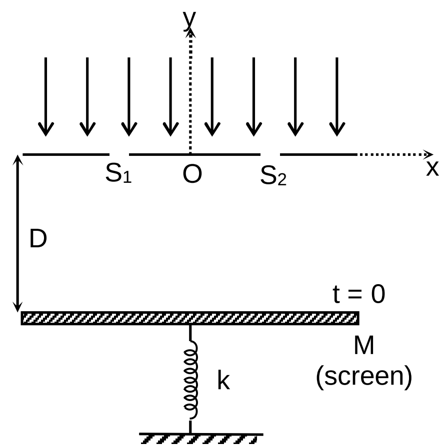
$N = Mg + Ma_0 + M\omega^2 x$

x = 0.1 meter

Question Stem for Question Nos. 18 and 19

Question Stem

A monochromatic light of wavelength 'λ' is incident on two slits S_1 and S_2 made on x-axis. The mid point of the line S_1S_2 is considered as origin and 'd' is the distance between the two slits which are symmetric about y-axis. The interference pattern is observed on a plate of mass M which is acting as a screen. The screen is attached to one end of a vertical spring of spring constant 'k' and the other end of spring is attached with the ground. Spring is massless and screen always remains in horizontal plane. The distance between the plane of slits and the screen is D (D>>d). Assuming S_1 and S_2 are coherent in nature. At t = 0, screen is released and spring is in its natural length. It is given that M = 1kg, k = 1 N/m. D = $\frac{5Mg}{k}$, g = 10 m/s^2, d = 200λ. The x co-ordinate (in cm) of the first order maxima on the screen as a function of time is given by $\pm(\lambda_1 - \lambda_2 \cos t)$.



18. The value of λ_1 is …………..

Ans. 30.00

19. The value of λ_2 is ……………

Ans. 5.00

Sol. (Q.18-19)

x-coordinate of first order maxima = $\pm\frac{\lambda D'}{d}$

Where $D' = D + \frac{Mg}{k}(1-\cos\omega t)$, $\left(\omega = \sqrt{\frac{k}{M}}\right)$

$= \frac{5Mg}{k} + \frac{Mg}{k}(1-\cos t) = 60 - 10\cos t$

So, $x = \pm\frac{1}{200}(60-10\cos t)\,\text{m} = \pm(30-5\cos t)$ cm

Chemistry PART – II

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

20. In which of the following crystals alternate tetrahedral voids are occupied

(A) NaCl

(B) ZnS

(C) CaF_2

(D) Na_2O

Ans. B

Sol. ZnS

21. The enthalpy of combustion of C and CO are – 393.5 kJ and – 283 kJ, respectively, the enthalpy of formation of CO is

(A) -110.5 kJ

(B) 6765 kJ

(C) -676.5 kJ

(D) 273.6 kJ

Ans. A

Sol.
$$C + O_2 \longrightarrow CO_2 \qquad \Delta H = -393.5 \text{ kJ} \qquad \dots (1)$$
$$CO + \frac{1}{2}O_2 \longrightarrow CO_2 \qquad \Delta H = -283.0 \text{ kJ} \qquad \dots (2)$$
$$C + \frac{1}{2}O_2 \longrightarrow CO \qquad \Delta H = -110.5 \text{ kJ}$$

22. $HNO_3 + P_2O_5 \longrightarrow A + B$

Product A and B are

(A) H_3PO_3, N_2O_3

(B) HPO_3, N_2O_5

(C) H_3PO_3, N_2O_4

(D) HPO_3, NO_2

Ans. B

Sol. $HNO_3 + P_2O_5 \longrightarrow HPO_3 + N_2O_5$

23. During electrolysis of aqueous $CuSO_4$, two Faraday of charge was passed. Find the volume of O_2 gas produced at anode at STP in litres?

(A) 5.6 L

(B) 11.2 L

(C) 22.4 L

(D) 2.8 L

Ans. B

Sol. $2H_2O \longrightarrow 4H^+ + O_2 + 4e^-$

Section – A (Maximum Marks: 24)

This section contains ***SIX (06)*** *question. Each question has* ***FOUR*** *options (A), (B), (C) and (D).* ***ONE OR MOER THAN ONE*** *of these four option(s) is (are) correct answer(s).*

24. Which of the following are correct statements?

(A) The hydro metallurgy process of extraction of silver metal is based on complex formation

(B) Cinnabar ore is concentrated by froth floatation process

(C) The process of converting hydrated alumina into alumina is called as calcination

(D) In alumino-thermite process Al is used as reducing agent

Ans. A, B, C, D

25. Which of the following is/are incorrect statements?

(A) Heating mixture of Cu_2O and Cu_2S give Cu

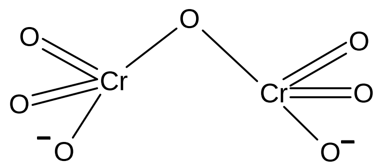
(B) Zn is the common metal in ore Brass, Bronze and German Silver

(C) Native silver form a complex with NaCN in presence of N_2

(D) In dichromate dianion all Cr – O bonds are equivalent

Ans. B, C, D

Sol. B = Cu
$C = O_2$ gas
D = Six Cr – O bonds are equivalent.



26. Which of the following cations cannot be separated by passing H_2S through their solutions to which NH_4Cl and NH_4OH have been added?

(A) Ca^{+2} and Ni^{+2}

(B) Mg^{+2} and Mn^{+2}

(C) Ni^{+2} and Co^{+2}

(D) Mn^{+2} and Zn^{+2}

Ans. C, D

Sol. NiS, CoS = black ppt.
MnS = buff clour ppt.
ZnS = white ppt.

27. Which of the following are correct regarding Xe and its compound?

(A) XeF_2 and SbF_5 reacts to form $[XeF][SbF_6]$

(B) XeF_6 and RbF reacts to form $[XeF_5][RbF_2]$

(C) XeF_6 and SiO_2 reacts to form $XeOF_4$ and SiF_4

(D) XeO_3 can be prepared by direct reaction between constituent elements

Ans. A, C

Sol.
$$XeF_2 + SbF_5 \longrightarrow [XeF][SbF_6]$$
$$XeF_6 + RbF \longrightarrow Rb[XeF_7]$$
$$XeF_6 + SiO_2 \longrightarrow XeOF_4 + SiF_4$$
$$XeF_6 + H_2O \longrightarrow XeO_3$$

28. Following are exothermic reactions

(A) Dehydrogenation of ethane to ethene

(B) Conversion of graphite to diamond

(C) Combustion of ethane

(D) Heat of neutralization

Ans. C, D

29. Which of the following is/are correct statement (s)?

(A) 100% tetrahedral voids are occupied by carbon in diamond

(B) Coordination number of HCP is 12

(C) In CsCl the coordination number of each ion is 8

(D) HCP shows ABCABCABC……..arrangement

Ans. B, C

Sol. 50% tetrahedral void in diamond
HCP = ABABAB…..

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. To a 56 ml H_2O_2 solution excess of acidified solution of KI was added. The I_2 liberated requires 200 ml of 0.3 N $Na_2S_2O_3$. Calculate the volume strength of H_2O_2.

Ans. 6

Sol. $56 \times N = 200 \times 0.3$

$N = \frac{60}{56}$

$\text{Volume strength} = \frac{60}{56} \times 5.6 = 6$

31. How many among the following metals are soluble in NH_3 / NH_4OH
Ag, Cu, Zn, Al , Ni, Fe, Pb

Ans. 4

Sol. Ag, Cu, Zn, Ni

32. 25 calories of heat is required to raise the temperature by $10^{o}C$ for 0.5 mole of a unknown gas at constant volume. Find molar heat capacity at constant pressure of unknown gas (in calorie). [Given R = 2 calorie/mole.K]

Ans. 7

Sol. $q = 25 \times 2 = 50 \text{ cal / mole}$

$C_V = \frac{50}{10} = 5 \text{ cal / mole / K}$

$C_P = 5 + 2 = 7 \text{ cal / mole / K}$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

All values are in Kcal per mole at $25^{o}C$ given below

$\Delta H^{o}_{Combustion(ethane)} = -372.0$

$\Delta H^{o}_{Combustion(propane)} = -530.0$

ΔH^{o} for $C(graphite) \longrightarrow C(g) = 172.0$

Bond energy of H – H = 104.0

ΔH^{o}_{f} of $H_2O(\ell) = -68.0$

ΔH^{o}_{f} of $CO_2(g) = -94.0$

33. Find the C – C bond energy in Kcal/mole_______

Ans. 82.00

34. Find the C – H bond energy in Kcal/mole______

Ans. 99.00

Sol. **(for Q. 33 and 34)**

$$C_2H_6 + \frac{7}{2}O_2 \longrightarrow 2CO_2 + 3H_2O$$
$$-372 = 2\times(-94) + 3(-68) - \Delta H^{o}_{f}(C_2H_6)$$
$$\Delta H^{o}_{f}(C_2H_6) = -20$$
$$C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$$
$$-530 = 3(-94) + 4(-68) - \Delta H^{o}_{f}(C_3H_8)$$
$$\Delta H^{o}_{f}(C_3H_8) = -24$$
$$2C(s) + 3H_2 \longrightarrow C_2H_6$$
$$-20 = 2\times 172 + 3\times(104) - (x + 6y)$$
$$3C(s) + 4H_2 \longrightarrow C_3H_8$$
$$-24 = 3\times 172 + 4(104) - (2x + 8y)$$

After solving, we get x and y

x = 82 Kcal/mol

y = 99 Kcal/mol

Question Stem for Question Nos. 35 and 36

Question Stem

7.02 gram of benzene containing 0.122 gram of benzoic acid freezes at $0.512^{o}C$ below the freezing point of pure benzene, benzoic acid exists as a dimer during this van't Hoff factor is found to be $x \times 10^{-1}$ and degree of association y%. (K_f for benezene is $5.12\ K/molal$).

35. The value of x is________

Ans. 7.02

36. The value of y is________

Ans. 59.60

Sol. **(for the Q. 35 and 36)**

$$\text{Molality} = \frac{1}{1000} \times \frac{1000}{7.02}$$

$$0.512 = i \times 5.12 \times \frac{1}{7.02}$$

$$i = 0.702 = 7.02 \times 10^{-1}$$

$$i = 0.702 = 1 - \frac{\alpha}{2}$$

$$\alpha = 59.60\%$$

Question Stem for Question Nos. 37 and 38

Question Stem

The equivalent conductivity of 0.02 N solution of monobasic acid is 15.75 $S\,cm^2\,eq^{-1}$. If equivalent conductivity of the acid at infinite dilution is $350\ S\,cm^2\,eq^{-1}$, the degree of dissociation found to be x% and it's Ka is found to be $y \times 10^{-5}$

37. The value of x is________

Ans. 4.50

38. The value of y is________

Ans. 4.05

Sol. **(for the Q. 37 and 38)**

$$\alpha = \frac{15.75}{350} = 0.045$$

= 4.5%

x = 4.50

$$K_a = \frac{C\alpha^2}{1-\alpha} \qquad 1-\alpha \cong 1$$

$$K_a = C\alpha^2$$

$$= (0.02)\left(4.5 \times 10^{-2}\right)^2$$

$$= 4.05 \times 10^{-5}$$

y = 4.05

Mathematics PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

39. The minimum value of $f(x) = \frac{(x-2)^{10} + (x-2)^{11} + 3(x-2)^{12} + (x-2)^{15} + (x-2)^{23} + (x-2)^{25}}{(x-2)^{15}}$ $\forall$ x > 2

(A) 2

(B) 4

(C) 6

(D) 8

Ans. D

Sol. $f(x) = (x-2)^{-5} + (x-2)^{-4} + 3(x-2)^{-3} + 1 + (x-2)^{8} + (x-2)^{10}$
Say $x - 2 = t\ (> 0)$
$f(x) = t^{-5} + t^{-4} + 3t^{-3} + 1 + t^{8} + t^{10}$. Now apply AM $\geq$ GM
$$\frac{t^{-5} + t^{-4} + t^{-3} + t^{-3} + t^{-3} + 1 + t^{8} + t^{10}}{8} \geq \left(t^{-5} \cdot t^{-4} \cdot t^{-3} \cdot t^{-3} \cdot t^{-3} \cdot 1 \cdot t^{8} \cdot t^{10}\right)^{\frac{1}{8}}$$
$\Rightarrow f(x) \geq 8\ \forall\ x > 2$ min value of f(x) is 8

40. The number of integral solution of the inequality $4 < x_1 + x_2 + x_3 + x_4 < 9$ where $x_1, x_2, x_3, x_4 \geq 0$

(A) 263

(B) 264

(C) 425

(D) none of these

Ans. C

Sol. $x_1 + x_2 + x_3 + x_4 < 9 \Rightarrow x_1 + x_2 + x_3 + x_4 + A = 10$
Number of integral solution = ${}^{12}C_4$ and $x_1 + x_2 + x_3 + x_4 \leq 4$
$x_1 + x_2 + x_3 + x_4 + B = 4$ number of integral solution ${}^{8}C_4$
So, total number of integral solution ${}^{12}C_4 - {}^{8}C_4 = 495 - 70 = 425$

41. A matrix of order 3×3 formed by using the elements {2021, 2022, ….. 2029} without repetition. Then the probability that first row of matrix has entries increasing order

(A) $\frac{1}{3}$

(B) $\frac{1}{4}$

(C) $\frac{1}{5}$

(D) $\frac{1}{6}$

Ans. D

Sol. Required probability = $\frac{\text{Favourable cases}}{\text{Total cases}} = \frac{{}^9C_3 \times 1 \times \underline{|6}}{\underline{|9}} = \frac{1}{6}$

42. Number of different terms in the expression $(1 + x)^{300} + (1 + x^2)^{200} + (1 + x^3)^{150}$ are

(A) 385

(B) 386

(C) 384

(D) 387

Ans. A

Sol. Let $A = (1 + x)^{300}$, $B = (1 + x^2)^{200}$, $C = (1 + x^3)^{150}$
$n(A) = 301$, $n(B) = 201$, $n(C) = 151$
$n(A \cap B) = 151$, $n(B \cap C) = 67$, $n(A \cap C) = 101$, $n(A \cap B \cap C) = 51$
$\therefore n(A \cup B \cup C) = 385$

Section – A (Maximum Marks: 24)

This section contains ***SIX (06)*** *question. Each question has* ***FOUR*** *options (A), (B), (C) and (D).* ***ONE OR MOER THAN ONE*** *of these four option(s) is (are) correct answer(s).*

43. Let z_1, z_2, z_3, z_4 are distinct complex number such that $|z_1| = |z_2| = |z_3| = |z_4|$.
If $|tz_1 + z_2 + z_3 + z_4| = |z_1 + tz_2 + z_3 + z_4| = |z_1 + z_2 + tz_3 + z_4|$ where $t \in R - \{1\}$, then

(A) z_1, z_2, z_3, z_4 must be the vertices of a rectangle

(B) z_1, z_2, z_3, z_4 may be the vertices of a square

(C) z_1, z_2, z_3, z_4 are collinear

(D) z_1, z_2, z_3, z_4 must be vertices of a rhombus

Ans. A, B

Sol. Let $z = z_1 + z_2 + z_3 + z_4$ then $|z - (1 - t)z_1| = |z - (1 - t)z_2| = |z - (1 - t)z_3|$
$\Rightarrow$ z is equidistant from 3 points so z will be the circumcentre of Δ formed by these points and distance of points $(1 - t)z_1$, $(1 - t)z_2$ and $(1 - t)z_3$ from (0, 0) are equal so $z = 0$
$\Rightarrow z_1 + z_2 + z_3 + z_4 = 0$ and $z_1 + z_2 = -z_3 - z_4 \Rightarrow z_1, z_2, -z_3, -z_4$ are vertices of a parallelogram described in a circle centre at (0, 0) and so a rectangle whose diagonal intersect at origin

44. Let A be a 4×4 matrix such that each entry of A is either 2 or –1. Let d = det(A), then

(A) d is divisible by 9

(B) d is always divisible by 27

(C) d is always divisible by 81

(D) d is composite number

Ans. A, B, D

Sol. Let $|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{vmatrix}$. If $R_1 \to R_1 - R_4$, $R_2 \to R_2 - R_4$, $R_3 \to R_3 - R_4$

Then elements of I, II, III row are 3, 0, or – 3
So, we take 3 common, then

$|A| = 3^3 \begin{vmatrix} \quad \end{vmatrix}$ $\Rightarrow$ |A| is divisible by 27

45. If three AM, A_1, A_2, A_3 and three HM H_1, H_2, H_3 are inserted between 3 and 6 and $A_1H_3 - A_2H_2$ is equal to r

(A) the value of r is equal to 2

(B) $\sum\limits_{n=r+1}^{100} \frac{1}{(2n+1)^2 - 1} = \frac{25}{101}$

(C) $\sum\limits_{n=r+1}^{\infty} \frac{n}{2^{n+1}} = 1$

(D) none of these

Ans. B, C

Sol. 3, A_1, A_2, A_3, 6 $\Rightarrow A_1 = 3 + \frac{3}{4} = \frac{15}{4}$

$A_2 = 3 + \frac{6}{4} = \frac{9}{2}$; $A_3 = 3 + \frac{9}{4} = \frac{21}{4}$ and $\frac{1}{3}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{6}$

$\Rightarrow H_1 = \frac{24}{7}, H_2 = 4$ and $H_3 = \frac{24}{5}$

Thus, $A_1H_3 - A_2H_2 = \frac{15}{4} \times \frac{24}{5} - \frac{9}{2} \times 4 = 0$

Now, $\sum\limits_{n=1}^{100} \frac{1}{(2n+1)^2 - 1}$; $T_n = \frac{1}{(2n+2)(2n)} = \frac{1}{4(n)(n+1)} = \frac{(n+1)-(n)}{4(n+1)(n)}$

$\frac{1}{4}\sum\limits_{n=1}^{100}\left(\frac{1}{n} - \frac{1}{n+1}\right) = \frac{1}{4}\left(1 - \frac{1}{101}\right) = \frac{25}{101}$ and $\sum\limits_{n=1}^{\infty} \frac{n}{2^{n+1}}$

$$T_n = \frac{n}{2\cdot 2^n} = \frac{1}{2}\left[\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+.....\right]$$

$$T_n = \frac{1}{2}S$$

$$S = \frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+.....\infty\ ;\ \frac{S}{2} = \frac{1}{2^2}+\frac{2}{2^3}+.....\infty$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2}+\frac{1}{2^2}+.....\infty\ ;\ \frac{S}{2} = \frac{\frac{1}{2}}{1-\frac{1}{2}} \Rightarrow S = 2$$

46. Let n be the number of ways in which the letters of the word “FIITJEE” can be arranged so that vowels appears at odd places and m be the number of ways in which “FIITJEE” can be arranged so that letters F, T, J appear in the same position as in the word “FIITJEE”, then correct statement is/are

(A) the value of n is 36

(B) the number of zero’s at the end in $\underline{|n}$ is 8

(C) the value of m + n is 42

(D) the value of n – m is 20

Ans. A, B, C

Sol. $n = {}^4C_4 \cdot \frac{\underline{|4}}{\underline{|2}\,\underline{|2}} \times {}^3C_3 \cdot \underline{|3} = 36$

Number of zero’s in $\underline{|36}$ = exponent of 5 in $\underline{|36} = 8$

$m = \frac{\underline{|4}}{\underline{|2}\,\underline{|2}} = 6$

47. Let $\alpha_1, \alpha_2, ... \alpha_{20}$ are roots of equation $1 + 2x + 3x^2 + ... + 11x^{10} + 10x^{11} + ... x^{20} = 0$, then

(A) equation has 10 repeated roots

(B) all roots are complex

(C) $|\alpha_1| + |\alpha_2| + ... + |\alpha_{20}| = 20$

(D) equation has 10 real roots

Ans. A, B, C

Sol. $1 + 2x + 3x^2 + ... + 11x^{10} + 10x^{11} + ... x^{20} = (1 + x + ... + x^{10})^2$

$$\Rightarrow \left(\frac{1-x^{11}}{1-x}\right)^2 = 0 \Rightarrow x = \cos\frac{2k\pi}{11} + i\sin\frac{2k\pi}{11},\ k = 1, 2, ... 10$$

48. Sum of the series $\left({}^{10}C_1\right)^2 + 2\left({}^{10}C_2\right)^2 + 3\left({}^{10}C_3\right)^2 + \ldots\ldots 10\left({}^{10}C_{10}\right)^2$ equals

(A) $10 \cdot {}^{20}C_{10}$

(B) $10 \cdot {}^{19}C_9$

(C) $\left(\frac{110}{21}\right) \times {}^{21}C_{11}$

(D) $5 \cdot {}^{20}C_{10}$

Ans. B, D

Sol. $S = 0\left({}^{10}C_{10}\right)^2 + 1\left({}^{10}C_1\right)^2 + 2\left({}^{10}C_2\right)^2 + \ldots\ldots 10\left({}^{10}C_{10}\right)^2$

$S = 10\left({}^{10}C_{10}\right)^2 + 9\left({}^{10}C_1\right)^2 + 8\left({}^{10}C_2\right)^2 + \ldots\ldots 0\left({}^{10}C_{10}\right)^2$

$2S = 10\left[\left({}^{10}C_0\right)^2 + \left({}^{10}C_1\right)^2 + \left({}^{10}C_2\right)^2 + \ldots\ldots \left({}^{10}C_{10}\right)^2\right]$

$2S = 10 \times {}^{20}C_{10} = 10 \times \frac{(20)!}{10!\,10!}$

$S = 10 \times \frac{19!}{9!\,10!}$; $S = 10 \cdot {}^{19}C_9$

Section – B (Maximum Marks: 12)

This section contains THREE (03) questions. The answer to each question is a NON-NEGATIVE INTEGER.

49. The number of real solution of equation $2021x^6 + 2021x^5 - 2020x^4 + x^3 + 2020x^2 + 2021x - 2021 = 0$

Ans. 2

Sol. $2021x^6 + 2021x^5 - 2020x^4 + x^3 + 2020x^2 + 2021x - 2021 = (x^2 + x - 1)(2021x^4 + x^2 + 2021) = 0$

50. The sum of last 100 digits of numbers 1.2.3.4 …..2021 + 2020 is

Ans. 4

Sol. 2021! + 2020 exponent of 5 in 2021! is $\left[\frac{2021}{5}\right] + \left[\frac{2021}{25}\right] + \left[\frac{2021}{625}\right] + \left[\frac{2021}{3125}\right]$ + …..

404 + 80 + 3 = 487

So, sum of last 100 digits is 2 + 0 + 0 + 2 = 4

51. Let z be a non-real complex number which satisfy the equation $z^{11} = 1$, then find $\sum_{k=1}^{10} \frac{1}{1 + z^{4k} + z^{8k}}$

Ans. 7

Sol. $\sum_{k=1}^{10}\frac{1}{1+z^{4k}+z^{8k}}=\sum_{k=1}^{10}\frac{z^{4k}-1}{z^{12k}-1}=\sum_{k=1}^{10}\frac{z^{4k}-1}{z^{k}-1}=\sum_{k=1}^{10}1+z^{k}+z^{2k}+z^{3k}$

$= 10 + (z + z^2 + \ldots\ z^{10}) + (z^2 + z^4 + z^6 + \ldots\ z^{20}) + (z^3 + z^6 + \ldots\ z^{30})$

$= 10 + (-1) + (-1) + (-1) = 7$

Section – C (Maximum Marks: 12)

This section contains ***THREE (03)*** *question stems. There are* ***TWO (02)*** *questions corresponding to each question stem. The answer to each question is a* ***NUMERICAL VALUE.*** *If the numerical value has more than two decimal places, truncate/round-off the value to* ***TWO*** *decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar unit vector such that the angle between every pair of them is $\frac{\pi}{3}$.

If $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$, where p, q, r are scalars; then

52. The value of $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^4$ is _____

Ans. 0.25

53. The value of $20p^2 + 21q^2 - 22r^2$ is _____

Ans. 9.50

Sol. (Q.52-53)

$\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$ taking dot product with $\vec{a}$

$0+[abc]=p\vec{a}\cdot\vec{a}+q\vec{a}\cdot\vec{b}+r\vec{a}\cdot\vec{c}$

$[abc] = p+\frac{q}{2}+\frac{r}{2}$ ….. (1)

Similarly taking dot product with $\vec{b}$

$0=\frac{p}{2}+q+\frac{r}{2}$ ….. (2)

and taking dot product with $\vec{c}$

$\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\frac{p}{2}+\frac{q}{2}+r$ ….. (3)

From equation (1) and (3), we get p = r

From equation (2) p = –q

$\Rightarrow [abc] = p-\frac{p}{2}+\frac{p}{2}=p$

But we know that $\left[\bar{a}\ \bar{b}\ \bar{c}\right]^2=\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c}\end{vmatrix}=\begin{vmatrix}1 & \frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & 1 & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2} & 1\end{vmatrix}$

$\Rightarrow \left[\bar{a}\ \bar{b}\ \bar{c}\right]^2=\frac{1}{2}$

$\Rightarrow \left[\vec{a}\ \vec{b}\ \vec{c}\right]^4 = \frac{1}{4} = 0.25$

$p = \frac{1}{\sqrt{2}}, q = -\frac{1}{\sqrt{2}}; r = \frac{1}{\sqrt{2}}$

Now, $20p^2 + 21q^2 - 22r^2 = \frac{20}{2} + \frac{21}{2} - \frac{22}{2} = 9.50$

Question Stem for Question Nos. 54 and 55

Question Stem

Let α, β be real numbers such that the system of linear equations

$$x + y + z = 2$$
$$x + 2y + 3z = 5$$
$$x + 3y + \alpha z = \beta$$

has infinitely many solutions. Let |M| represent the determinant of the matrix

$$M = \begin{bmatrix} -2 & -3+\beta & \sin\theta - 2 \\ 1 & \sin\theta + 2 & \gamma \\ \alpha & 2\sin\theta - 1 & -\sin\theta + 2 + 2\gamma \end{bmatrix}$$

where $\gamma \in R$ and $\theta \in [0, 2\pi]$ and min value of determinant M is 8 and foot of perpendicular from point P(α – 4, β – 10, 1) (where (α, β) for which the above system of linear equation has infinitely many solution) to the plane x + 2y – 2z – 10 = 0 is F (a, b, c), then

54. The value of $\frac{\gamma}{5}$ is _____

Ans. 0.20

55. The value of $\frac{3}{2}(a+b+c)$ is _____

Ans. 2.50

Sol. (Q.54-55)

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & \alpha \end{vmatrix} = 0 \Rightarrow \alpha = 5$$

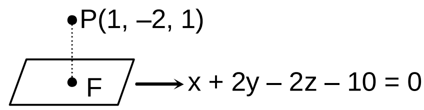
$$D_1 = \begin{vmatrix} 2 & 1 & 1 \\ 5 & 2 & 3 \\ \beta & 3 & 5 \end{vmatrix} = 0 \Rightarrow \beta = 8 \text{ and } M = \begin{vmatrix} -2 & 5 & \sin\theta - 2 \\ 1 & \sin\theta + 2 & \gamma \\ 5 & 2\sin\theta - 1 & -\sin\theta + 2 + 2\gamma \end{vmatrix} = 0$$

Applying $R_3 \to R_3 - 2R_2 + R_1$

$$|M| = \begin{vmatrix} -2 & 5 & \sin\theta - 2 \\ 1 & \sin\theta + 2 & \gamma \\ 1 & 0 & 0 \end{vmatrix} = 5\gamma - (\sin^2\theta - 4)$$

Minimum value of det |m| is obtained when $\sin^2\theta = 1$

$\Rightarrow 5\gamma + 3 = 8 \Rightarrow \gamma = 1 \Rightarrow \frac{\gamma}{5} = 0.20$ and point P $\equiv$ (α – 4, β – 10, 1) $\equiv$ (1, –2, 1)



Foot of perpendicular F is $\frac{x-1}{1}=\frac{y+2}{2}=\frac{z-1}{-2}=\frac{-(1-4-2-10)}{1^2+2^2+(-2)^2}$ = $\left(\frac{8}{3},\frac{4}{3},-\frac{7}{3}\right)$

Then, $\frac{3}{2}(a+b+c)=2.50$

Question Stem for Question Nos. 56 and 57

Question Stem

There are four parcels namely A, B, C and D in a room. A particular I-phone is in one of them. The probability that it is in parcel A is $\frac{2}{5}$, but if it is not in parcel A its probability is equally likely to be either in B or C or D.

I sent any delivery man to bring one of a parcel to me. The Probability that my delivery man brings the parcel A is $\frac{2}{3}$ and the probability that he brings any other parcel is $\frac{1}{3}$. Let α = I get the particular I-phone, β = my delivery man brings parcel A

56. The value of $\frac{15}{2}P(\alpha)$ _____

Ans. 3.50

57. The value of $7P\left(\frac{\beta}{\alpha}\right)$ _____

Ans. 4.00

Sol. (Q.56-57)

$P(A)=\frac{2}{5}$; $P(B)=P(C)=P(D)=\frac{1}{5}$

Let F denote “I will got I-Phone”

Then, $P(\alpha)=P(A)\cdot P\left(\frac{F}{A}\right)+P(\text{not in A})P\left(\frac{F}{\text{not in A}}\right)$ = $\frac{4}{15}+\frac{3}{15}=\frac{7}{15}$ and $P\left(\frac{\beta}{\alpha}\right)=\frac{\frac{4}{15}}{\frac{4}{15}+\frac{3}{15}}=\frac{4}{7}$

Hence, $\frac{15}{2}P(\alpha)=\frac{15}{2}\times\frac{7}{15}=3.50$ and $7P\left(\frac{\beta}{\alpha}\right)=4.00$